

Sergey Gendel

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Summary

Machine Learning Engineer expert with 10 (ten) + years of international hands-on experience in developing software solutions for complex ML problems in various fields, including health care, metrology, aerodynamics, signal and image processing. Experience in building agent systems and shipping to production.

Areas of Expertise

Programming

- Python (NumPy Scikit-learn XGBoost Pandas, SciPy)
- MATLAB
- R

Frameworks

- Pytorch
- Keras

ML Skills

- Classical ML & DL Tools
- Reinforcement Learning
- LLM, RAG, LoRA, Lang Chain
- GAN

Shell scripting

- Unix/Linux
- Bash

Databases

- MySQL
- MongoDB

Platforms

- Hugging Face
- Weights & Biases
- Docker

Cloud Computing

- AWS (SageMaker, S3, EC2, Jupyter)

Version Control

- Git

Management Tools

- JIRA, Confluence

Experience

November 2018 – July 2024

Senior Machine Learning Engineer

Samsung R&D Center, Israel

- Developed and tested automated systems for flash memory performance improvement. Implemented an AI-based algorithm for up to 50% memory error reduction.
- Researched novel mathematical and ML methods for flash memory lifetime extension.
- Automatization of flash memory programming scheme.
- Designed, developed, and implemented AI - based systems and data pipelines for the Samsung Memory business, evaluated their performance, and implemented them in simulation.

- Enhanced existing software solutions to improve flash memory reliability using state-of-the-art AI methods. Collaborated with the HW development team to integrate the novel AI methods for use in flash memory.

February 2016 – October 2018

Algorithm Specialist

Nova Measuring Instruments Ltd, Israel

- Designed, developed and implemented machine learning algorithms for dimensional and materials metrology solutions for advanced process control used in semiconductor manufacturing.
- Researched mathematical and statistical models for semiconductor metrology applied science problems solutions
- Developed and tested automated system control for semiconductor manufacturing.
- Worked closely with customers to plan, design and coordinate the development of software solutions.

February 2014 – January 2016

AI Engineer

Optimata Ltd, Israel

- Developed biomathematical, statistical, and computational machine learning models for personalized therapy of oncology patients.
- Designed and prototyped machine learning algorithms for cancer progression and treatment effectiveness prediction.
- Collaborated closely with medical professionals, hospitals and medical clinics to determine the project's design, and physical and mathematical parameters.
- Worked with the team to create and deploy well-engineered, reliable and robust code for various applications.

December 2015 – May 2016

Machine Learning Consulting

Gili Medical Ltd., Israel

- Created predictive machine learning algorithms aimed at forecasting blood sugar levels and detecting early signs of hypoglycemia shocks, leveraging data collected from the device's non-invasive measurements.
- Utilized advanced techniques including SVM, Decision Trees, Neural Networks, Hidden Markov Models, CNN, LSTM, and RNN to analyze device data and enhance prediction accuracy.
- Contributed to the development of life-saving technology by improving the accuracy of hypoglycemia shocks prediction, supporting the company's mission to advance non-invasive blood sugar monitoring solutions.

2004 – 2012

Teaching Assistant • Technion

Israel Institute of Technology

- Conducted tutorials, graded assignments, and provided individual consultations for courses in Partial Differential Equations, Regular Differential Equations, and Numerical Analysis.

Education

2014

Technion -The Israel Institute of Technology

Ph.D. in Mechanical Engineering

2007

Technion -The Israel Institute of Technology

M.Sc. in Applied Mathematics

2000

Technion -The Israel Institute of Technology

B.Sc. in Applied Mathematics

Patents

DE-NOISING USING MULTIPLE THRESHOLD-EXPERT MACHINE LEARNING MODELS

<https://patents.justia.com/patent/20220293192>

ERROR DETECTION AND CORRECTION USING MACHINE LEARNING

<https://patents.justia.com/patent/20220013189>

Publications

Kleiman, A., Gendel, S., Degani, D., & Gottlieb, O. (2016). Self-excited dynamics of an elastically restrained slender rigid body in uniform compressible laminar flow. *Nonlinear Dynamics*, 86(4), 2207–2230

Gendel, S., Gottlieb, O., & Degani, D. (2015). Fluid–Structure Interaction of an Elastically Mounted Slender Body at High Incidence. *AIAA Journal*, 53(5), 1309–1318.

Gendel, S., Gottlieb, O., & Degani, D. (2013, June). Flow around an elastically mounted slender body at high incidence. *21st AIAA Computational Fluid Dynamics Conference*.

Gendel, S., Gottlieb, O. & Degani, D. (2012) Nonlinear spatiotemporal fluid-structure interaction of an elastically mounted slender body at high incidence, *Proc. 33rd IACMM Symposium on Computational Mechanics*

Gendel, S., Degani, D. & Gottlieb O. (2010, June) Numerical analysis of the influence of finite rotations on vortex-induced vibrations of a tethered cylinder undergoing internal resonances, *Proc. 6th IUTAM Symp. on Bluff Body Wakes and Vortex-induced Vibrations, BBVIV-6*, 329-332.